

Homework (Habanero)

- Q1.** A patient needs $2x + 3y \geq 12$ units of medicine. If $x = 2$, what is the minimum value of y ?
- (A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q2.** Graph the inequality $y < 2x - 5$. What type of line should be used for the boundary?
- (A) Solid
(B) Dashed
(C) Dotted
(D) Double-solid
(E) None
- Q3.** A cell grows at a rate of $y = 3x - 2$. If the cell count is greater than 10, what inequality represents this situation?
- (A) $y > 10$
(B) $y < 10$
(C) $y \geq 10$
(D) $y \leq 10$
(E) $y = 10$
- Q4.** Solve the inequality $x + 2y \leq 8$. What is the y -intercept?
- (A) 0
(B) 2
(C) 4
(D) 6
(E) 8
- Q5.** The inequality $y > \frac{1}{2}x - 3$ represents a region in the coordinate plane. What is the equation of the boundary line?
- (A) $y = \frac{1}{2}x - 3$
(B) $y = -\frac{1}{2}x + 3$
(C) $y = \frac{1}{2}x + 3$
(D) $y = -\frac{1}{2}x - 3$
(E) $y = 2x - 3$
- Q6.** What type of line should be used to graph the inequality $2x + 5y \geq 10$?
- (A) Solid
(B) Dashed
(C) Dotted
(D) Double-solid
(E) None
- Q7.** The region defined by $y < 2x + 1$ includes the point $(0, 0)$. What is the relationship between x and y at this point?
- (A) $y < x$
(B) $y > x$
(C) $y = x$
(D) $y < 2x$
(E) $y > 2x$
- Q8.** Graph the inequality $x - 2y > 4$. What is the x -intercept of the boundary line?
- (A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Open Ended Questions (FRQ)

- Q9.** Solve the inequality $3x + 2y \leq 12$. Graph the solution region and identify the vertices of the region. Write your answer in the space provided below.
- Q10.** A doctor needs to mix two medicines, x and y , in a ratio of 2 : 3. If the total amount of medicine is greater than 20 units, write an inequality to represent this situation. Solve the inequality and graph the solution region.

Chef's Tips

- Tip 1: Emphasize the importance of using the correct type of line when graphing inequalities.
- Tip 2: Encourage students to check their work by plugging in values from the solution region.
- Tip 3: Remind students to label the axes and include a title when graphing inequalities.